

LEE YING SHENG

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Education

B.S Electrical & Computer Engineering — Shanghai Jiao Tong University (CGPA: 3.26)

09/22 – Present

- Minors: Data Science and Computer Science
- Relevant Coursework: Computational Methods for Data Science, Introduction to Robotics, Data Structures and Algorithms
- Activities: SJTU Asia Pacific Study @ Nanyang Technological University Singapore Exchange Program

Work Experience

Teaching Assistant — Shanghai Jiao Tong University

02/24 – 12/25

- Taught / Advised close to 300 students in Academic Writing, Supply Chain Management, and Engineering courses.
- Cooperated with 2 or 3 other Teaching Assistants to organized student competitions within courses.
- Received positive feedback from close to 80% of the class in Teaching Assistant Evaluations.

Frontend Machine Learning — Moham Labs

10/24 – 02/25

- Developed a UI/UX for an agentic AI chatbot system with C# .NET WPF and XAML, integrating it into an existing prototype.
- Collaborated with Graphic Designers and UI/UX specialists when building the frontend of the prototype.
- Architected agentic workflows with LangGraph to automate BIM tasks and introduce AI-assisted functions in Autodesk Revit.

Projects

SJTU International Events Platform — LaowaiCentral

04/26 – Present

- Architected an aggregation engine with WeChat RSS to centralize event data from 50+ Official Accounts.
- Engineered a pipeline to filter spam, categorize content, and generate multilingual summaries using LLM prompt engineering.
- Developed a newsletter system via Aliyun Direct Mail that bypasses strict ISP spam filters.
- Integrated an analytics suite to track user retention, click-through rates (CTR), and newsletter conversion metrics.

SVM Gradient Descent Evaluation in Fake News Detection — Shanghai Jiao Tong University

11/25 – 12/25

- Collaborated in a 3-person team to engineer 3 custom Gradient Descent optimizers for SVM without built-in libraries.
- Processed 45K+ data samples via TF-IDF feature engineering, achieving a 97.17% weighted accuracy in fake news detection.
- Built a Shiny app for interactive parameter exploration and real-time model experimentation.

RISC-V Pipeline Processor Design — Shanghai Jiao Tong University

05/25 – 08/25

- Engineered a 5-stage RISC-V pipelined processor in Verilog HDL, with data forwarding and stalling logic to resolve hazards.
- Implemented a memory system with cache logic and virtual-to-physical address mapping.
- Executed full-stack hardware modeling in Xilinx Vivado, from RTL coding to functional simulation and verification.

Roomba Path Planning Simulation — Shanghai Jiao Tong University

10/24 – 12/24

- Developed a 2D simulation framework to evaluate and compare multiple autonomous Roomba path-planning algorithms.
- Conducted multi-iteration simulations across diverse room layouts and dynamic obstacle scenarios.
- Engineered an evaluation matrix to analyze algorithm efficiency across four dimensions: code complexity, cleaning coverage percentage, completion speed, and environmental adaptability.
- Responsible for algorithm development, system integration, and performance benchmarking as part of a 3-member team.

Amphibious Tracked Vehicle — Shanghai Jiao Tong University

09/23 – 08/24

- Awarded 1st Prize in China's 13th National Marine Vehicle Design Competition, ranking top among 230 regional champions.
- Designed and prototyped a cross-domain platform capable of transitioning between deep water and land using a dual-propulsion system (propeller and tracks).
- Utilized SolidWorks to model a hydrodynamic hull and a 10cm retractable track mechanism to minimize underwater drag.
- Implemented long-range wireless communication (up to 100m) using the nRF24L01 module with complex steering logic.

Skills

- Programing Languages: Python, R Programming Language, C++, C, C#, TypeScript, SQL, Verilog, Assembly
- Frameworks and Systems: Django, React, React Native, LangGraph, .NET WPF, Next.js, Git, Arduino, Postgres
- AI Tools: Cursor, Claude Code, Prompt Engineering, Context Management, Token Management
- Design: Solidworks, 3D Printing, Laser Cutting
- Languages: English, Chinese, Malay, Cantonese